

Lab 1

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Deadline: 2022-09-25 23:59:59, Beijing time

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Attachment: Your java files such as Exercise02_01.java and Exercise02_02.java

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Exercise01_04.java

1.4 (Print a table) Write a program that displays the following table:

a	a^2	a^3	a^4
1	1	1	1
2	4	8	16
3	9	27	81
4	16	64	256

Exercise02_01.java

2.1 (Convert mile to kilometer) Write a program that reads a mile in a double value from the console, converts it to kilometers, and displays the result. The formula for the conversion is as follows:

```
1 mile = 1.6 kilometers
```

Here is a sample run:

```
Enter miles: 96 [Enter]  
96 miles is 153.6 kilometers
```

Exercise02_24.java

2.24 (Convert Celsius to Fahrenheit) Write a program that reads a Celsius degree in a double value from the console, then converts it to Fahrenheit, and displays the result. The formula for the conversion is as follows:

$$\text{fahrenheit} = (9/5) * \text{Celsius} + 32$$

Here is a sample run:

```
Enter a temperature in Celsius: 43.5
43.5 Celsius is 110.3 Fahrenheit
```

Hint: in Java, 9 / 5 is 1, but 9.0 / 5 is 1.8

Exercise02_02.java

2.2 (Compute the volume of a triangle) Write a program that reads in the length of sides of an equilateral triangle and computes the area and volume using the following formulas:

$$\text{Area} = \frac{\sqrt{3}}{4} (\text{length of sides})^2$$

$$\text{volume} = \text{area} * \text{height}$$

Here is a sample run:

```
Enter length of the sides and height of the equilateral triangle: 3 5 [Enter]
The area is 3.897
The volume of the triangular prism is 19.486
```

Hint: `System.out.printf("%.3f", volume);` and `Math.sqrt(3)` can be used.

Exercise02_06.java

2.6 (Multiply the digits in an integer) Write a program that reads an integer between 100 and 999 and multiplies all the digits in the integer. For example, if an integer is 132, the multiplication of all its digits is 6.

Hint: Use the % operator to extract digits, and use the / operator to remove the extracted digit. For instance, $932 \% 10 = 2$ and $932 / 10 = 93$.

Here is a sample run:

```
Enter an integer between 100 and 1000: 999 [Enter]
The multiplication of all digits in 999 is 729
```

Exercise02_09.java

2.9 (Physics: acceleration) Average acceleration is defined as the change of velocity divided by the time taken to make the change, as given by the following formula:

$$a = \frac{v_1 - v_0}{t}$$

Write down a program that prompts the user to enter the starting velocity v_0 in meters/second, the ending velocity v_1 in meters/second, and the time span t in seconds, then display the average acceleration. Here is the sample run:

```
Enter v0, v1 and t: 5.5 50.9 4.5 [Enter]
The average acceleration is 10.0889
```

Exercise02_10.java

2.10 (Science: calculating energy) Write a program that calculates the energy needed to heat water from an initial temperature to a final temperature. Your program should prompt the user to enter the amount of water in kilograms and the initial and final temperature of the water. The formula to compute the energy is

$$Q = M * (\text{finalTemperature} - \text{initialTemperature}) * 4184$$

Where M is the weight of water in kilograms, initial and final temperatures are in degree Celsius, and energy Q is in joules. Here is a sample run:

```
Enter the amount of water in kilograms: 55.5
Enter the initial temperature: 3.5
Enter the final temperature: 10.5
The energy needed is 1625484.0
```

Exercise02_12.java

2.12 (Physics: finding runway length) Given an airplane's acceleration a and take-off speed v , you can compute the minimum runway length needed for an airplane to take off using the following formula:

$$\text{length} = \frac{v^2}{2a}$$

Write a program that prompts the user to enter v in meters/second (m/s) and the acceleration a in meters/second squared (m/s^2), then, displays the minimum runway length. Here is a sample run:

```
Enter speed and acceleration: 60 3.5 [Enter]
The minimum runway length for this airplane is 514.286 meters.
```